

Stochastic Model of Local Fluctuating Magnetic Fields in Pulsed NMR

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Introduction to Pulsed NMR

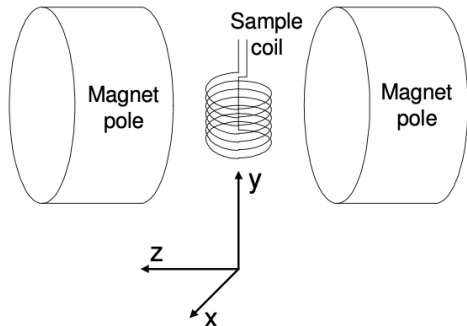


Figure: Magnet and coil set-up [TeachSpin Pulsed/CW NMR Manual]

Physical picture

- Place a sample in a strong magnetic field and briefly hit it with a radiofrequency (RF) pulse.
- After the pulse, the sample emits a decaying oscillatory signal.

Question:

What causes the decay, and what information about the system does it encode?

Spin precession, magnetization, and RF pulses

Spins behave like tiny magnets

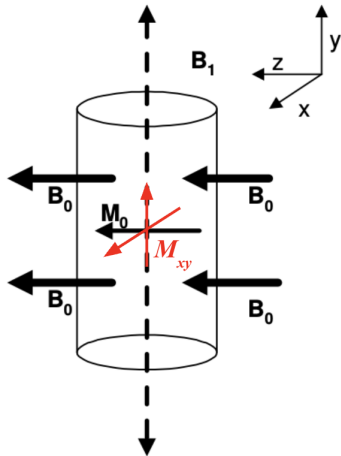
- In a magnetic field

$$\omega_0 = \gamma B_0$$

- Net magnetization M_0 aligns along z in thermal equilibrium

RF pulse creates a measurable signal

- M_0 is tipped onto xy-plane
- Spins precess coherently $\rightarrow$ oscillating signal



Decay of the transverse magnetization

The transverse magnetization will decay exponentially:

$$M_{xy}(t) = M_0 e^{-t/T_2}$$

T_2 – Spin-spin relaxation time

The timescale for loss of phase coherence...

Interpretation:

Decay = loss of phase coherence
among spins

Question:

What causes this loss of coherence, and
how can we model it?

Phase decoherence

The signal could decay via two mechanisms:

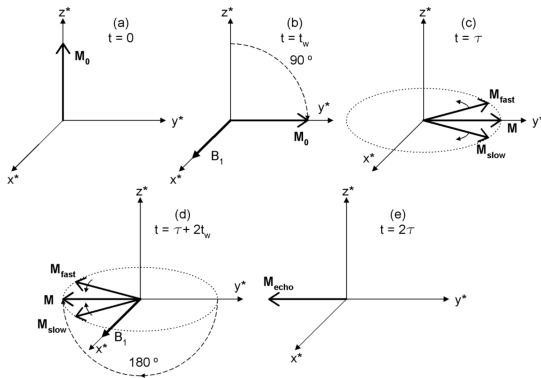
Field inhomogeneity – the static field varies across the sample (*reversible*)

Time-dependent fluctuations – fields change in time due to molecular dynamics (*irreversible*)

Both result in a *distribution* of precession frequencies → loss of phase coherence.

Spin echo

We can indeed reverse the effect of field inhomogeneity



⚡ Echo amplitude $<$ initial signal

Loss of information must arise from time-dependent (stochastic) fluctuations

Phase accumulation

Each spin accumulates a phase

$$\phi(t) = \gamma \int_0^t \mathfrak{B}(t') dt'$$

- If $\mathfrak{B}$ is static, the phase evolution is deterministic
- If $\mathfrak{B}(t)$ fluctuates, the phase becomes random

The signal is thus given by

$$S(t) = \sum_{k=1}^N e^{i\phi_k(t)} \implies S(t) = \langle e^{i\phi(t)} \rangle$$

Gaussianity of the stochastic field

Say $\mathfrak{B}(t) = \sum_i^N b_i(t)$, where the b_i are

zero-mean

weakly correlated

of the same order

Then for large N ,

$$\mathfrak{B}(t) \sim \mathcal{N}(0, \Delta^2)$$

This is insufficient for a full characterization; indeed, we need

$$C(T) = \langle \mathfrak{B}(t) \mathfrak{B}(t + T) \rangle$$

– the autocorrelation function of the field

Field correlations

$\phi(t) \sim \mathcal{N}(0, \langle \phi^2(t) \rangle)$ as a linear functional of a Gaussian process

$$\langle e^{i\phi(t)} \rangle = e^{-\frac{1}{2} \langle \phi^2(t) \rangle}$$

Phase variance is determined entirely by the time correlations of the field:

$$\langle \phi^2(t) \rangle = \gamma^2 \int_0^t \int_0^t \langle \mathfrak{B}(t_1) \mathfrak{B}(t_2) \rangle dt_1 dt_2$$

What correlation structure of $\mathfrak{B}(t)$ produces

$$\langle \phi^2(t) \rangle \propto t \quad \implies \quad M_{xy}(t) \sim e^{-t}?$$

Modeling the fluctuating field

Ornstein-Uhlenbeck process

Describes a field that randomly fluctuates with amplitude σ , but relaxes back to zero over a timescale τ_c

$$d\mathfrak{B}_t = -\frac{1}{\tau_c}\mathfrak{B}_t dt + \sigma dW_t$$

- $dW_t \sim \mathcal{N}(0, dt)$ – Gaussian white noise

Correlation function

$$C(T) = \langle \mathfrak{B}(t)\mathfrak{B}(t+T) \rangle = \Delta^2 e^{-|T|/\tau_c}$$

Phase variance

$$\langle \phi^2(t) \rangle \sim \begin{cases} t^2, & t \ll \tau_c \\ t, & t \gg \tau_c \end{cases} \implies \boxed{M_{xy}(t) \sim e^{-t}, \quad t \gg \tau_c}$$

Simulating the OU process

Under an isotropicity assumption, there is a mapping $(T_1, T_2) \mapsto (\tau_c, \Delta^2)$:

$$\underbrace{\tau_c = \frac{1}{\omega_0} \sqrt{2 \left(\frac{T_1}{T_2} - 1 \right)}}_{\text{memory time of fluctuations}}, \quad \underbrace{\Delta^2 = \frac{1}{4\gamma^2 T_1 \tau_c} (1 + \omega_0^2 \tau_c^2)}_{\text{fluctuation amplitude}}$$

We can feed these parameters into

$$\phi_{\text{echo}}(2\tau) = \gamma \left(\int_0^\tau \mathfrak{B}_z(t) dt' - \int_\tau^{2\tau} \mathfrak{B}_z(t) dt' \right)$$

and predict the echo amplitude via

$$S(2\tau) \propto \left| \left\langle e^{i\phi(2\tau)} \right\rangle \right|$$

Experimental objectives

Goal

Test whether OU-derived parameters from T_1 , T_2 predict spin echo decay

- Extract τ_c, Δ^2 from T_1, T_2
- Predict echo amplitude
- Compare with experiment

Samples probe different correlation-time regimes

Sample	Dynamics	τ_c Regime	Purpose
Light Mineral Oil	Fast	$\tau_c \ll 1/\omega_0$	Validation
Heavy Mineral Oil	Intermediate	$\tau_c \sim 1/\omega_0$	Tests limits
Vaseline (optional)	Slow	$\tau_c \gg 1/\omega_0$	Failure case

Magnetization dynamics (T_1)

Longitudinal magnetization relaxes exponentially to equilibrium. Assuming $M(0) = 0$,

$$M_z(t) = M_0(1 - e^{-t/T_1})$$

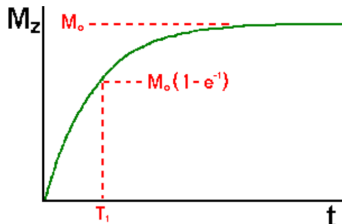


Figure: Magnetization vs. time for a sample placed in a magnetic field [TeachSpin Pulsed/CW NMR Manual]

T_1 – Spin-lattice relaxation time

Characteristic timescale of relaxation toward thermal equilibrium, caused by exchange of

energy

angular momentum

from nuclei to lattice

Measuring T_1 : inversion recovery

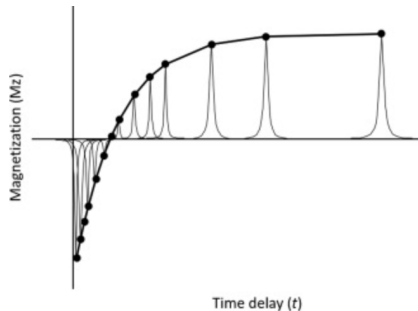
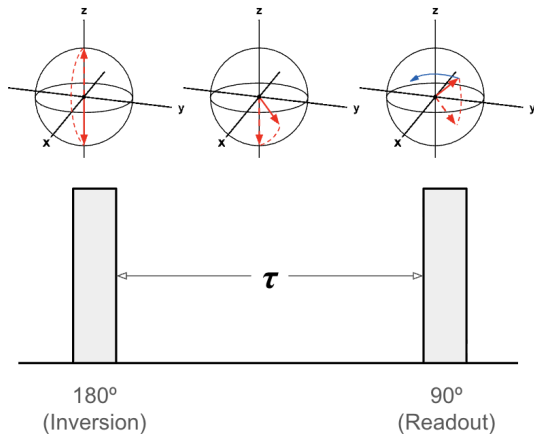


Figure: [ScienceDirect – Inversion Recovery]

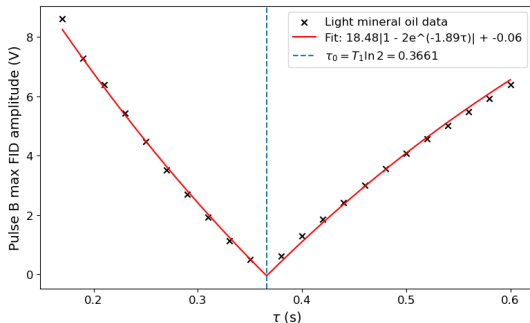
$$M_z(t) = M_0(1 - 2e^{-t/T_1})$$

T_1 measurements

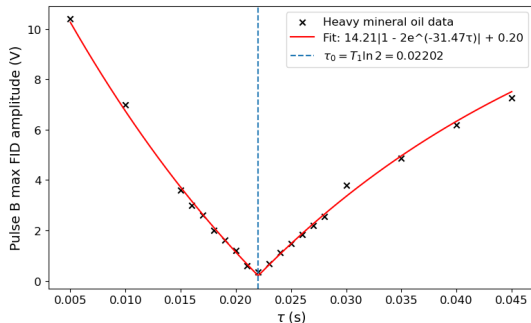
Env. Out gives only $S(\tau) \propto |M_z(\tau)|$, so we adjust the fit:

$$S(\tau) = A \left| 1 - 2e^{-\tau/T_1} \right| + C$$

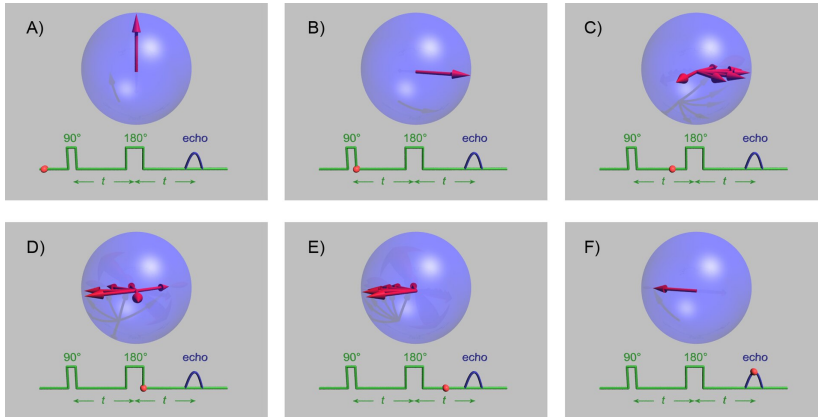
$$T_{1,\text{light}} = 528 \pm 1 \text{ ms}$$



$$T_{1,\text{heavy}} = 31.8 \pm 0.1 \text{ ms}$$



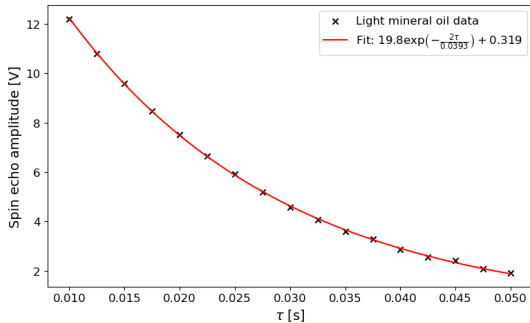
Measuring T_2 : spin echo decay



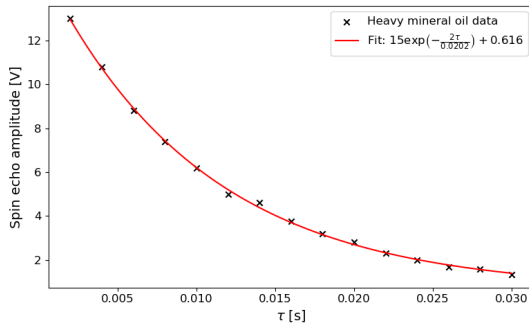
$$M_{xy}(2\tau) = M_0 e^{-2\tau/T_2} \implies S(2\tau) = A e^{-2\tau/T_2} + C$$

T_2 measurements

$$T_{2,\text{light}} = 39.3 \pm 0.5 \text{ ms}$$



$$T_{2,\text{heavy}} = 20.2 \pm 0.5 \text{ ms}$$



Simulation parameters

Sample	T_1 (ms)	T_2 (ms)	f_{rec} (MHz)
Light Oil	528	39.3	21.12730
Heavy Oil	31.8	20.2	21.02330

⇓

Sample	τ_c (ns)	Δ^2 (G ²)	σ (T · s ^{-1/2})
Light Oil	37.6	0.45	0.49
Heavy Oil	8.09	2.91	2.68

Table: Extraction of OU parameters from measured relaxation times.

Simulated OU field and autocorrelation

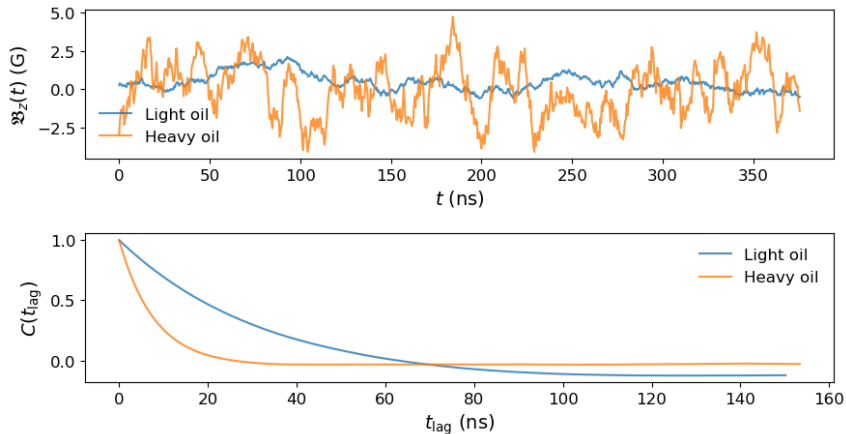
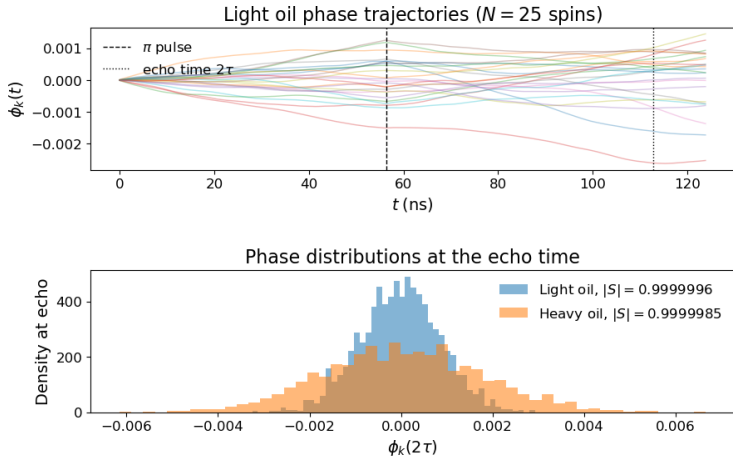


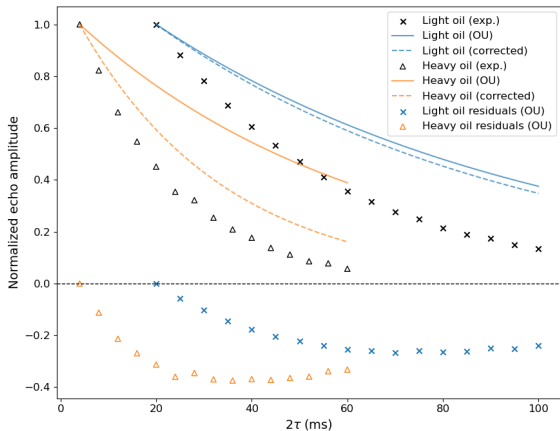
Figure: OU field $\mathfrak{B}_z(t)$ realization and autocorrelation $C(t_{\text{lag}})$ over 2000 realizations

Simulated phase accumulation

For illustration purposes, we pick $\tau = \tau_{c,\text{light}}$, and $|S(2\tau)| = |\langle \exp i\phi(2\tau) \rangle|$



Experimental v.s. simulated echo decay



Potential failure modes:

- Account for the effect of $\mathfrak{B}_{xy}$ (energy relaxation),
- Real liquids have multiple motional process

$$C(t) = \sum_i \Delta_i^2 \exp(-|t|/\tau_i),$$

- Molecular diffusion through gradients

$$S(2\tau) \sim \exp(-D\gamma G^2 \tau^3),$$

- Circular validation: test on a different dataset.

Conclusion

Key result

The OU model *qualitatively* predicts spin echo decay, but systematically underestimates relaxation.

Regime dependence

Heavy oil: strong deviation

Light oil: better agreement

Interpretation

Extend model: multi-timescale correlations or include transverse fluctuations